



GLENWOOD SYSTEMS – TECHNICAL INTERVIEW PANEL CHEAT SHEET

Java & SQL Technical Interview Question Bank

Instructions to Interviewers

- Ask **ONE question** from each section.
- Each section carries **1 Mark**.
- Probe further if required.
- Evaluate Concept Clarity, Confidence, Logical Thinking and Explanation.

SECTION 1: JAVA FUNDAMENTALS

(Data Types, Variables, Operators, Control Statements)

1. What is the difference between JDK, JRE and JVM?
2. Why is Java called Platform Independent?
3. Explain Primitive and Non-Primitive Data Types with examples.
4. What is the difference between == operator and equals() method?
5. What are implicit and explicit type casting?
6. Explain break, continue and return statements.
7. What is the difference between while loop and do-while loop?
8. What are command-line arguments in Java?
9. Explain the switch statement. Can it work with String?
10. What is the main() method? Why is it declared as public static void?

SECTION 2: OBJECT ORIENTED PROGRAMMING (OOPs)

1. What are the four pillars of OOP?
2. What is the difference between Class and Object?
3. Explain Inheritance with a real-life example.
4. What is Method Overloading?
5. What is Method Overriding?
6. Difference between Overloading and Overriding.
7. Explain Encapsulation.



8. What is Abstraction?
 9. Difference between Abstract Class and Interface.
 10. What is Polymorphism? Explain compile-time and run-time polymorphism.
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SECTION 3: EXCEPTION HANDLING & PACKAGES

1. What is an Exception?
 2. Difference between Checked and Unchecked Exceptions.
 3. Explain try, catch and finally blocks.
 4. Difference between throw and throws.
 5. Can finally block be skipped?
 6. What is Exception Propagation?
 7. How do you create a Custom Exception?
 8. What is the purpose of packages?
 9. Difference between import package.* and import specific class.
 10. What happens if an exception is not handled?
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SECTION 4: COLLECTIONS & STRINGS

1. What is the Java Collections Framework?
 2. Difference between Array and ArrayList.
 3. Difference between List, Set and Map.
 4. Explain HashMap.
 5. Difference between HashMap and Hashtable.
 6. Difference between String, StringBuffer and StringBuilder.
 7. Why are Strings Immutable?
 8. Explain HashSet.
 9. Difference between Comparable and Comparator.
 10. How does HashMap internally work?
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SECTION 5: METHODS, ARRAYS & PROGRAMMING LOGIC

1. Difference between Parameters and Arguments.
2. Explain Pass by Value in Java.
3. How do you reverse an array?



4. Write logic to find the largest element in an array.
 5. Difference between 1D and 2D arrays.
 6. How do you remove duplicate elements?
 7. Explain Recursion.
 8. Write logic to check whether a number is Prime.
 9. Write logic to check Palindrome.
 10. Write logic to generate Fibonacci Series.
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SECTION 6: SQL FUNDAMENTALS

1. What is DBMS?
 2. Difference between DBMS and RDBMS.
 3. What is SQL?
 4. Difference between DDL, DML, DCL and TCL.
 5. What is a Primary Key?
 6. What is a Foreign Key?
 7. Explain NOT NULL, UNIQUE and CHECK constraints.
 8. Difference between DELETE, TRUNCATE and DROP.
 9. What is Normalization?
 10. Explain different Normal Forms.
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SECTION 7: SQL QUERY WRITING

1. Write a query to display all records.
 2. Difference between WHERE and HAVING.
 3. Explain ORDER BY.
 4. What is DISTINCT?
 5. Explain GROUP BY.
 6. Difference between COUNT(*) and COUNT(column).
 7. How do you retrieve the top N records?
 8. Explain LIKE operator.
 9. Difference between IN and EXISTS.
 10. Write a query to display duplicate records.
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SECTION 8: JOINS & RELATIONSHIPS

1. What is a Join?
2. Difference between INNER JOIN and LEFT JOIN.
3. Explain RIGHT JOIN.
4. Explain FULL OUTER JOIN.
5. Difference between SELF JOIN and CROSS JOIN.

6. When would you use SELF JOIN?
 7. Difference between Primary Key and Foreign Key.
 8. Explain Referential Integrity.
 9. Can two tables be joined without a foreign key?
 10. Explain Many-to-Many Relationships.
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SECTION 9: AGGREGATE FUNCTIONS & SUBQUERIES

1. Explain COUNT(), SUM(), AVG(), MAX() and MIN().
 2. What is a Subquery?
 3. Difference between Correlated and Non-Correlated Subqueries.
 4. What is EXISTS?
 5. Difference between EXISTS and IN.
 6. Write a query using GROUP BY and HAVING.
 7. What is a Scalar Subquery?
 8. Explain Nested Queries.
 9. How do Aggregate Functions handle NULL values?
 10. Difference between UNION and UNION ALL.
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SECTION 10: PRACTICAL PROBLEM SOLVING (JAVA & SQL)

1. Write a Java program to reverse a String.
 2. Write a Java program to check whether a number is Prime.
 3. Write a Java program to find the second largest number in an array.
 4. Write a Java program to count vowels in a String.
 5. Write a Java program to remove duplicate elements from an array.
 6. Write an SQL query to find the second highest salary.
 7. Write an SQL query to retrieve duplicate employee records.
 8. Write an SQL query to display employees working in more than one project.
 9. Write an SQL query using INNER JOIN on Employee and Department tables.
 10. Given a small business scenario, write an appropriate SQL query and explain your approach.
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